THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 7 & 8: **Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A constant force of 8 N displaces a body by 25 m in the direction of the force. The work done is: 1

(a) 0.32 J **(b)** 400 J **(c)** 200 J **(d)** 33 J

2. The kinetic energy of a body of mass 1000 kg moving at 10 m s^{-1} is: 1

(a) 50000 J **(b)** 100000 J **(c)** 10000 J **(d)** 25000 J

3. The potential energy of a 4 kg body at a height of 20 m (g = 10 m s^{-2}) is: 1

(a) 400 J **(b)** 80 J **(c)** 1600 J **(d)** 800 J

4. A machine does 900 J of work in 3 s. Its power is: 1

(a) 300 W **(b)** 600 W **(c)** 2700 W **(d)** 0 W

5. A ramp of length 12 m raises a load to a height of 3 m (ignore friction). Its mechanical advantage is: 1

(a) 8 **(b)** 4 **(c)** 5 **(d)** 0.25

6. A sound wave of frequency 4 Hz has a time period of: 1

(a) 4 s **(b)** 0.25 s **(c)** 0.12 s **(d)** 0.5 s

7. A source of frequency 170 Hz produces waves of wavelength 2 m in a medium. The speed of the wave is: 1

(a) 340 m s^{-1} **(b)** 85 m s^{-1} **(c)** 172 m s^{-1} **(d)** 680 m s^{-1}

8. Sound of frequency 170 Hz travels at 340 m s^{-1}. Its wavelength is: 1

(a) 0.5 m **(b)** 57800 m **(c)** 2 m **(d)** 4 m

9. An echo is heard 2 s after a shout. If the speed of sound is 340 m s^{-1}, the distance of the reflecting surface is: 1

(a) 350 m **(b)** 680 m **(c)** 340 m **(d)** 170 m

10. A sonar signal returns from the ocean floor after 2 s. If the speed in water is 1530 m s^{-1}, the depth is: 1

(a) 765 m **(b)** 3060 m **(c)** 1540 m **(d)** 1530 m

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Define mechanical advantage. 2

12. What is the use of a fixed pulley? 2

13. Define reverberation. 2

14. Name two uses of ultrasound. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why are roads on hills made to wind around? 3

16. Describe how a microphone and a speaker work. 3

17. Classify the levers: scissors, wheelbarrow and tweezers. 3

OR

On a seesaw a 15 kg child sits 2 m from the fulcrum. Where must a 30 kg child sit to balance it? 3

18. In a movie, a spacecraft explosion in space shows a flash of light along with sound. What is wrong? 3

OR

A source produces a wave of wavelength 3.44 m at a speed of 344 m s^{-1}. Find the time period. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A truck of 10 000 kg moving at 72 km h^{-1} enters an escape ramp inclined at 30°, with sand exerting 50 000 N. 5

(i) Find its speed in m s^{-1}.
(ii) Find its initial KE.
(iii) Find the length of the ramp it needs.
(iv) Why are escape ramps made of sand?

OR

(i) List the different forms of energy with one example each. 5
(ii) Explain energy conversion in a hydroelectric plant.
(iii) What is the gharat or panchakki?

20. (i) Describe the structure of the human ear and the path of sound. 5
(ii) Why do we have two ears?

OR

(i) Compare the speed of sound in air, water and steel (344, 1500, 5000 m s^{-1}). 5
(ii) Find the ratios water/air and steel/water.
(iii) Explain the trend.

END OF QUESTION PAPER — **Total: 40 Marks**

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